

Ziao Yang

+1 (781) 392-8208 | <https://yangziao56.github.io/homepage> | ziaoyang@brandeis.edu

EDUCATION

Brandeis University

Ph.D. Candidate in Computer Science

Advisor: Professor Hongfu Liu

Waltham, MA

09/2023 - 05/2027 (Expected)

Sun Yat-Sen University

B.Eng. in Computer Science and Technology

Guangzhou, China

09/2018 - 06/2022

RESEARCH INTERESTS

Large Language Models: Data Curation for Pre-Training and Post-Training, Reinforcement Learning and LLM Evaluation

Data-Centric Learning: Data Valuation, Influence Functions, and Data-Efficient Training

Spatiotemporal Forecasting: Transformer-based Video Prediction and Precipitation Nowcasting

RESEARCH & INDUSTRY EXPERIENCE

Adobe Research

Research Scientist Intern, Mentor: Dr. Vlad Morariu

Seattle, WA

05/2026 - 08/2026

Researched methods to reduce LLM inference latency for HTML generation in an Adobe product while preserving output quality.

- Developed and evaluated model distillation and parallel generation methods to improve the latency-quality trade-off.
- Built an end-to-end pipeline from synthetic data generation with a 480B teacher to fine-tuning, serving, and product evaluation of 8B student LLMs.

Google Tunix Research Collaboration

Academic Research Collaborator

Remote

09/2025 - 05/2026

Researched data-centric LLM post-training using Google Tunix on TPUs.

- Developed self-influence and outlier-based methods to filter low-signal samples in GRPO and DPO.
- Implemented DeepScaleR math-reasoning post-training and AIME 2024 evaluation on multi-host TPUs.

Adobe

Machine Learning Engineer Intern, Mentor: Dr. Lei Zhang

San Jose, CA

05/2025 - 08/2025

Developed controllable headline and email generation for Adobe GenStudio for Performance Marketing, supporting enterprise marketing workflows.

- Improved offline quality through multi-persona generation and evaluation.
- Trained a reward model and applied PPO for brand-aligned generation.

Y Combinator China

Intern, Mentor: Dr. Qi Lu

Beijing, China

03/2023 - 07/2023

Conducted AI industry research and startup scouting, translating technical and market trends into strategic insights for leadership.

- Prepared presentations and financing materials for Dr. Qi Lu, including “My Foundation Model Worldview.”
- Scouted emerging AI startups and synthesized insights from discussions with scientists at leading AI labs.

International Digital Economy Academy (IDEA)

Research Intern, Mentor: Dr. Jiaying Zhang

Shenzhen, China

03/2022 - 07/2022

Contributed to the Fengshenbang language-model project (4K+ GitHub stars).

- Pre-trained BERT (1.3B) and GPT-2 (3.5B) models from scratch.
- Built datasets of 27 million labeled examples for multi-task classification.
- Performed supervised continued pre-training across classification tasks using MAML and Reptile.

**RESEARCH &
INDUSTRY
EXPERIENCE**

Sun Yat-Sen University

Research Assistant, Mentor: Dr. Qifeng Lin

Guangzhou, China

2020 - 2022

Researched Transformer-based video and spatiotemporal forecasting.

- Developed TCTN, combining Transformer attention with 3D temporal convolutions to model short- and long-range dependencies for future-frame prediction on MovingMNIST and KTH.
- Extended TCTN to PTCT for autoregressive radar-echo forecasting and precipitation nowcasting on Radar Echo Guangzhou and HKO-7.

SKILLS

LLM Training: Post-Training (PPO, GRPO), Preference Optimization (DPO), Pre-Training from Scratch, Fine-Tuning, Distillation, Synthetic Data Generation

Tools: Python, PyTorch, JAX, Tmux, DeepSpeed, vLLM, SGLang, Hugging Face, Distributed Training (Multi-GPU, Multi-Host TPU)

PUBLICATIONS

Ziao Yang, Han Yue, Jian Chen, and Hongfu Liu. “Revisit, Extend, and Enhance Hessian-Free Influence Functions.” *Transactions on Machine Learning Research (TMLR)*, 2026.

Ziao Yang, Zizhang Chen, Lei Zhang, and Hongfu Liu. “Recontextualizing Famous Quotes for Brand Slogan Generation.” *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2026.

Ziao Yang, Longbo Huang, and Hongfu Liu. “Layer-Aware Influence for Online Data Valuation Estimation.” Submitted to *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026.

Ziao Yang, Han Yue, Jian Chen, and Hongfu Liu. “On the Inflation of KNN-Shapley Value.” Submitted to *Data-centric Machine Learning Research (DMLR)*, 2026.

Yinan Shen, **Ziao Yang**, and Hongfu Liu. “Validation-Induced Shapley Shifts: How Validation Structure Distorts Data Valuation.” *arXiv preprint arXiv:2607.03675*, 2026.

Ziao Yang, Xiangrui Yang, and Qifeng Lin. “TCTN: A 3D-Temporal Convolutional Transformer Network for Spatiotemporal Predictive Learning.” *arXiv preprint arXiv:2112.01085v1*, 2021.

Ziao Yang, Xiangrui Yang, and Qifeng Lin. “PTCT: Patches with 3D-Temporal Convolutional Transformer Network for Precipitation Nowcasting.” *arXiv preprint arXiv:2112.01085v2*, 2022.

**HONORS &
AWARDS**

Tianchi Big Data Competition - AI Helps Weather Forecast (Top 1%) 2022
Alibaba Group

National Level of College Student Innovation Competition (Top 1) 2022
Sun Yat-Sen University

TCTN Official Implementation on GitHub 2022

**PROFESSIONAL
ACTIVITIES**

Reviewer

Neural Information Processing Systems (**NeurIPS**) 2024 - Present

International Conference on Learning Representations (**ICLR**) 2024 - Present

International Conference on Machine Learning (**ICML**) 2024 - Present